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Comparative Study on Calculating CPU Burst Time using Different Machine Learning Algorithms

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***Abstract*—This paper gives a thorough analysis of works on the subject of expert assessment of CPU burst time using various machine learning algorithms. Knowing how long the CPU bursts for the processes will last is necessary for some CPU scheduling algorithms like SJF and SRTF to function. In particular, the non-preemptive SJF scheduling algorithm estimates the process that will be performed by the CPU in the least amount of burst time. One effective way of predicting CPU burst duration is an ML-based algorithm that estimates the burst-time of the processes. Throughout the study, we discovered that the effectiveness of different machine-learning approaches relies on the applications to which they are put. Our examination of the literature not only argues that these methods are competitive with conventional estimators on a single data set, but also demonstrates that they are responsive to the training data.**

***Index Terms*— CPU burst time, scheduling algorithms, SJF, ML algorithms.**

I. INTRODUCTION

The mechanism for granting access to tasks, processes, threads, or data flow is called a scheduling algorithm. The requirement to execute numerous processes simultaneously has prompted the creation of scheduling algorithms in today's multiprogramming and modern systems. There are several scheduling ways suggested to decide which tasks should be executed on priority basis. Some algorithms are preferred over others including Round Robin, SJF, and FCFS. This comparison's goal is to evaluate whether a method is better suited to certain tasks in the ready queue. Various algorithms, like FCFS, Shortest Job First, and others, which depend on time frame parameters like Arrival Time, Burst Time, Waiting Time, etc., are used to execute numerous processes in an optimised manner. The magnitude of the CPU burst must be known in order to employ SJF scheduling or SRTF. Burst time is how long a process requires to execute, expressed in milliseconds. The burst time accounts for the CPU time used by a process. The time spent doing I/O is not taken into account. SJF and SRTF algorithms require processes which have short duration bursts over those with longer ones to make sure that the smaller process will finish and exit the system faster than the bigger one. It is not simple to determine the next CPU burst size for a process; it requires additional work and computations.

Earlier, the length of CPU bursts was measured using mathematical formulas. That traditional way is called "Exponential Averaging (EA)", this technique uses the previous history of CPU bursts to predict the next burst. When multi-level platforms from various places in computational grids are employed to deliver a variety of services to a shared objective; work scheduling is seen as a crucial element in the infrastructure of

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a computational grid. Grid scheduling is the process of allocating resources across several administrative domains to Grid jobs. Our research concentrates on the many ML approaches that are employed and can be used for efficient estimation of CPU-burst time.

II. LITERATURE SURVEY

A Pourali, Amir Rahmani [1] A variety of time-stamped data, in particular CPU-burst data series, may be forecasted with a certain amount of accuracy by using expert machines like fuzzy systems. Some perks of fuzzy logic involve improved adaptability and optimal speed. The significance of the suggested technique is its capacity to anticipate a process's forthcoming CPU burst time based on previous information.

Tarek H, Sadam Al-Azani, Omar Obaidellah [2] By utilising ML techniques such as SVM, DT, ANN, and K-NN, the report recommends a technique to evaluate the CPU burst duration for activities. The "GWA-T-4 Auver Grid" grid workload was implemented in the suggested method to apply the attribute feature selection approaches choose the most solution that will work. depending on the outcomes of the experiment. Despite a strong link between the business factors and the burst Computational time, K-Nearest Neighbour consistently outperforms further Machine Learning algorithms on the grounds of RAE & CC.

Putera, Andysah, Utama, Siahaan [3] Calculations using three algorithms show the varying average waiting times. For a short while, first come first served is preferable. Shortest Job First is preferable when multiple processes are entering the CPU simultaneously. The best way to change the needed average waiting time is to use the Round Robin approach. Greater SJF or FCFS values will be established at round robin response time. All algorithms work well, although the rate of process advancement depends on the CPU load.

Sirisha B, Suresh K, Diya D S, Soumya [4] The suggested remedy employs a number of approaches, including k-nearest neighbours, decision learning, and linear regression, to determine the Central Processing Unit burst length for the method in the ready list. This strategy is more effective than earlier CPU scheduling techniques like SJF and SRTF.

J R Indusree, B Prabadevi [5] The experiment results clearly show that the proposed algorithm offers improved performance. In comparison to the RR algorithm, there are a lot fewer context switch. Additionally, it shortens turnaround and average waiting times.

Bushra Hamid, Inayat Ur-Rehman [6] This research suggests that the running performance of diverse scheduling algorithms with the recently proposed scheduling approach. In comparison to round robin and pre-emptive SJF, the proposed approach has less context switching, according to the comparison results. Application of the suggested method can effectively maximise system performance by reducing context switching.

Mahesh Kumar M.R, Renuka Rajendra.B, Niranjan C.K, Sreenath .M [7] The operation that has a shorter burst time than the preceding ones is chosen from the ready queue by a conventional SJF scheduling algorithm. This can be resumed as long as the final operation in the waiting list is completed. This technique always requires the CPU to execute the next operation with their burst length. This information isn't always accessible, though.

Dr Vandana [8] This study aims to update the EAF (Exponential Average Formula) utilising fuzzy systems. Prior to a process's execution, the suggested method's major goal is to predict and estimate its execution time. In order to obtain high flexibility and the desired calculation speed, fuzzy inference engines are used.

Ms. Vaibhavi Pandya, Mr. Indra Jeet Rajput [9] For the SRR scheduling, the execution's time quantum is determined beforehand at a specific time, however in this suggested scheduling. Although time quantum's algorithm is not predefined, it is based on freshly established work sets, modified. As a result, the approach does not employ conventional technique, so is more efficient than a static round robin.

Parul Prakram, Mohit Prakram [10] SJF is among the top CPU scheduling algorithms. Despite this, the CPU scheduling algorithm cannot be used in real world due to the difficulty in determining the precise time spent during a process's execution before it has truly executed. Consequently, utilising the suggested Fuzzy based utilising an SJF scheduling technique that has to be aware of approaching CPU-burst times for processes. It will be realised in a working computer system.

III. HELPFUL HINTS

A. Figures and Tables

TABLE I. COMPARISON TABLE OF ALL PAPERS

SR.NO	TITLE	JOURNAL NAME	YEAR	ALGORITHM USED	DESCRIPTION
1.	CPU Burst Time using Machine Learning Algorithm's [3]	International Journal of Advanced Research in Science, Communication and Technology (IJARSCT) [3]	2022	K-Nearest Neighbour, Decision learning, and Linear Regression.	It has been suggested how to figure out how long a CPU burst should last for each task. It uses a variety of approaches, including k-nearest neighbour, decision learning, and linear regression. This strategy is more effective than earlier CPU scheduling techniques like SJF and SRTF.
2.	A Machine Learning-Based Approach to Estimate the CPU-Burst Time for Processes in the Computational Grids [2]	3rd International Conference on Artificial Intelligence, Modeling and Simulation [2]	2015	Feature Selection CPU Scheduling Algorithm	In this research, they have proposed a method for predicting the duration of CPU burst using Machine Learning methods. Additionally, in order to choose the most crucial properties, the "GWA-T-4 Auver Grid" grid burden was put through attribute feature selection procedures. The best Machine Learning method is K-NN in practically every case when concerned with associated coefficient and Error.
3.	Prediction of Length of the Next CPU Burst in SJF Scheduling Algorithm using Dual Simplex Method [7]	2nd International Conference on Current Trends in Engineering and Technology, ICCTET'14 [7]	2014	Conventional SJF Scheduling algorithm, Machine Learning	In this situation, the standard Shortest Job First selects the operation from the waiting list where the CPU burst duration is minimum. This strategy always needs the CPU to run the next operation with their execution times until achieving the finish we practically can't estimate the upcoming CPU burst time.
4.	A Fuzzy-based Scheduling Algorithm for prediction of next CPU-burst time to implement Shortest Process Next [1]	International Association of Computer Science and Information Technology - Spring Conference [1]	2009	Fuzzy systems, Shortest Process Next (SPN)	A variety of time-stamped data, in particular CPU-burst data series, may be forecasted with a certain amount of accuracy by using expert machines like fuzzy systems. Some perks of fuzzy logic involve improved adaptability and optimal speed. The significance of the suggested technique is its capacity to anticipate a process's forthcoming CPU burst time based on previous information.
5.	A Fuzzy Built Methodology towards Predicting the Next CPU-Burst for SJF Scheduling Algorithm [10]	THINK INDIA JOURNAL [10]	2019	CPU Scheduling algorithms, SJF algorithm,	A lower bound "a," an upper bound "b," and a parameter "m" are used to create a triangular linear model, which is what the method in this case does. For the same objective, they used the Fuzzy constructed algorithm in order to calculate the exponential average Algorithm.

6.	Revised Formula for Estimating CPU-Burst [8]	Journal of International Academy of Physical Sciences [8]	2018	Average Exponential Formula (AEF), Fuzzy Inference System (FIS)	They used a set of processes that were randomly chosen, and they computed the results to forecast the CPU. Using MATLAB and C++, the updated exponential average formula was compared to the original EAF. Additionally, they attempted to present us with graphic comparisons using three distinct numbers for α, such as 0, 1, and 12.
7.	Enhanced round robin CPU scheduling with burst time-based time quantum [5]	IOP Conference Series: Materials Science and Engineering [5]	2017	FCFS algorithm, Priority scheduling algorithm	The experiment's findings unmistakably demonstrate that the proposed method provides better performance. In comparison to the RR algorithm, there are a lot fewer context switch. Additionally, it shortens turnaround and average waiting times.
8.	Performance Improvisation in Scheduling through Optimal Resource Utilization [9]	International Journal for Technological Research in Engineering [9]	2017	Dynamic time, quantum time quantum context switching	In terms of execution, the adaptive round robin scheduling is an effective scheduling technique. The suggested system's primary emphasis is on creating task sets and subsequently use the algorithm for adaptive round robin
9.	Comparison Analysis of CPU Scheduling: FCFS, SJF and Round Robin [4]	IJEDR | Volume 4, Issue 3 [4]	2016	SJF, Round Robin and CPU scheduling	Here are the outcomes of three methods that serve as illustrations of the varied mean waiting times. For small periods, the FCFS functions effectively. The SJF is better if the operations train and develop the CPU simultaneously. For altering the needed mean waiting times, the RR technique works well. RR quantum frequency will set a greater value for SJF or FCFS. All algorithms operate as intended, but CPU load controls how fast a process proceeds.
10.	An Optimized Shortest job first Scheduling Algorithm for CPU Scheduling [6]	Journal of Applied Environmental and Biological Sciences [6]	2015	Hybrid Scheduling algorithm, Preemptive SJF	The hybrid scheduling algorithm was created by them. The shortest job first technique plus the constraint on residual burst length of the executing process method make up this algorithm. The rationale behind such a strategy is that it starts this procedure on the task with the shortest burst time at launch. They also noted that the scheduling technique is proactive. As a result, when a new job joins, it can decide if it wants to preempt an active process.

B. Comparison Graphs

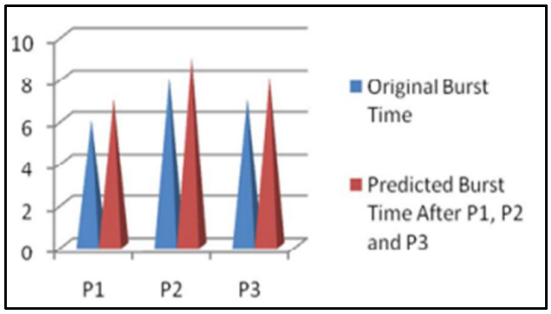


Figure 1. Analogy of actual burst time and predicted burst time using ML algorithms

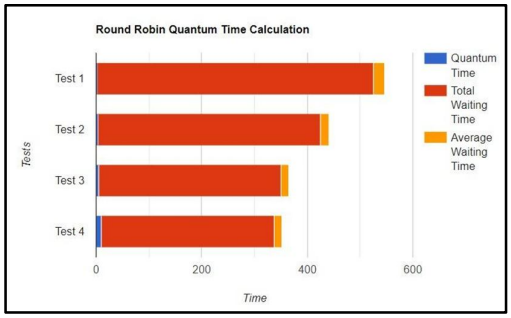


Figure 3. Accuracy Rate for ML Algorithms

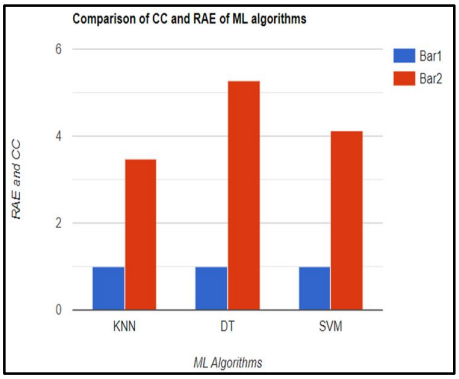


Figure 2. Comparison of Coefficient Comparison and Relative Absolute Error

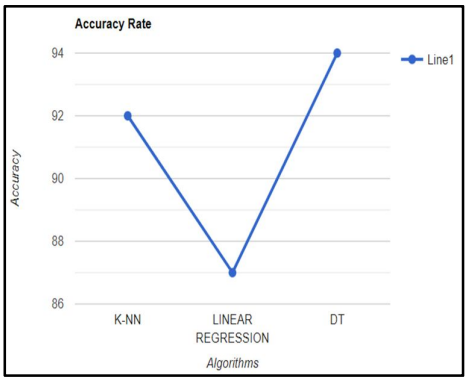


Figure 4. Plotting Burst time and Arrival Time using Scatter Plot

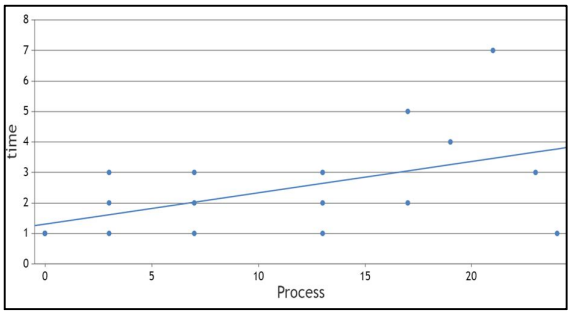


Figure 5. Round Robin Quantum Time Calculation

IV. Scope of Implementation

In this paper, further advanced and improved algorithms and building models to have a detailed comparative analysis of the CPU burst time dataset can be discussed. More features can be included for comparing algorithms such as accuracy, F1 score etc. In further research, more advanced ML models to predict the CPU burst time can be .

V. Conclusion

Discussing the different Machine-Learning Techniques utilised for CPU Burst time estimate is the major contribution of this review. As a conclusion it was found that in order to determine the optimal method to calculate CPU burst time for processes, a variety of techniques including linear regression, logistic regression,

decision trees, and KNN should be used. We should employ the approach that best fits the model while keeping in mind its constraints and the primary goal of enhancing its performance and effectiveness.
Our research also supports the idea that there is no single machine learning method that is ideal. Because of this, a greater understanding of the applicability and scope of a number of the strategies mentioned is urgently needed.

REFERENCES

[1] Pourali, Abdolghader, and Amir Masoud Rahmani. "A Fuzzy-based Scheduling Algorithm for prediction of next CPU-burst time to implement Shortest Process Next," International Association of Computer Science and Information Technology-Spring Conference. IEEE, 2009.

[2] Helmy, Tarek, Sadam Al-Azani, and Omar Bin-Obaidellah. "A machine learning-based approach to estimate the CPU-burst time for processes in the computational grids" 3rd International Conference on Artificial Intelligence, Modelling and Simulation (AIMS). IEEE, 2015.

[3] Samal, Prathamesh, Sagar Jha, and Raman Kumar Goyal. "CPU Burst-Time Estimation using Machine Learning," IEEE Delhi Section Conference (DELCON). IEEE, 2022.

[4] Siahaan, Andysah Putera Utama. "Comparison analysis of CPU scheduling: FCFS, SJF and Round Robin," International Journal of Engineering Development and Research 4.3 (2016): 124-132.

[5] Indusree, J. R., and B. Prabadevi. "Enhanced round robin CPU scheduling with burst time based time quantum," IOP Conference Series: Materials Science and Engineering. Vol. 263. No. 4. IOP Publishing, 2017.

[6] Hamayun, Maryam, and Hira Khurshid. "An optimized shortest job first scheduling algorithm for CPU scheduling," J. Appl. Environ. Biol. Sci 5.12 (2015): 42-46.

[7] Kumar, MR Mahesh, et al. "Prediction of length of the next CPU burst in SJF scheduling algorithm using dual simplex method," Second International Conference on Current Trends in Engineering and Technology-ICCTET 2014. IEEE, 2014.

[8] Vandana "Revised Formula for Estimating CPU-Burst," Journal of International Academy of Physical Sciences Vol. 22 No. 4 (2018)

[9] Ms.Vaibhavi Pandya, Mr.Indr Jeet Rajput, "PERFORMANCE IMPROVISATION IN SCHEDULING THROUGH OPTIMAL RESORCE UTILIZATION," International Journal For Technological Research In Engineering Volume 4, Issue 11, July-2017

[10] Prakram, Parul, and Mohit Prakram. "A Fuzzy Built Methodology towards Predicting the Next CPU-Burst For SJF Scheduling Algorithm," Think India Journal 22.17 (2019): 3851-3859.